

Belief propagation for diffusion scores on Markov chains: exact Gaussian baseline and the cost of Gaussian message closure — summary of results —

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1 Setting and exact identities

The clean signal is a Markov chain $a_{1:n}$, observed through coordinatewise Ornstein–Uhlenbeck noising at diffusion time $t > 0$:

$$x_i = \alpha_t a_i + \sqrt{\Delta_t} z_i, \quad \alpha_t = e^{-t}, \quad \Delta_t = 1 - e^{-2t}, \quad z_i \sim \mathcal{N}(0, 1) \text{ i.i.d.}$$

Two structural facts organize everything:

1. **The posterior is a chain.** A Markov prior plus coordinatewise noise gives a posterior $p_t(a_{1:n} | x_{1:n})$ whose factor graph is a chain, hence a tree, hence sum-product BP computes the posterior marginals *exactly*.
2. **The denoiser gives the score.** For every Markov prior,

$$s_i(x, t) = \partial_{x_i} \log p_t(x) = -\frac{x_i - \alpha_t m_i(x, t)}{\Delta_t}, \quad m_i(x, t) = \mathbb{E}[a_i | x_{1:n}].$$

So exactness of BP is never the issue on a chain. The issue is *representation*: for a continuous non-Gaussian chain, each BP message is a whole function of a continuous variable. Only in special cases (Gaussian: two scalars; finite alphabet: a finite vector) do the messages close in a finite-dimensional family.

2 Part I: the Gaussian AR(1) baseline is fully solvable

For the stationary Gaussian AR(1) prior ($\Sigma_0(i, j) = \rho^{|i-j|}$), BP messages stay Gaussian, and the score is also available directly by linear algebra, $s(x, t) = -\Sigma_t^{-1}x$ with $\Sigma_t = \alpha_t^2 \Sigma_0 + \Delta_t I$. Running both routes ($n = 40$, $\rho = 0.7$, $t = 0.8$):

$$\max_i |s_i^{\text{BP}} - s_i^{\text{matrix}}| \approx 6.7 \times 10^{-16},$$

i.e. agreement at machine precision, matching the algebraic proof of equivalence. This case serves as the exactly-solvable control for everything below.

3 Part II: general chains and the two algorithms

For non-Gaussian chains we compare two implementations of the same BP equations:

- **Reference grid BP.** Messages represented on a grid of M points, integrals by trapezoidal quadrature. Converged in M , this is numerically exact.
- **Gaussian projected BP** (assumed-density filtering style). After each exact message update, the outgoing message is replaced by the Gaussian with the same first two moments. This isolates precisely the error caused by *Gaussian message closure* — the approximation implicitly made when one treats a non-Gaussian problem with Gaussian/AMP-style machinery.

Test model: AR(1) chain with **Laplace innovations** (same correlation structure as the Gaussian case, $\rho = 0.85$, unit marginal variance, $n = 50$, heavier tails).

Is the reference trustworthy?

Two independent checks from the heavy runs:

- *Gaussian control:* grid BP vs the exact matrix score, 48 trials per t , grid sizes $M \in \{101, 201, 401\}$: relative score error $\sim 10^{-16}$ – 10^{-14} for all diffusion times.
- *Laplace grid convergence:* $M = 401$ vs $M = 801$, 16 trials per t : relative score difference $\leq 6 \times 10^{-6}$ at all times — three to five orders of magnitude below the closure error being measured.

4 Main result: the price of Gaussian message closure

Figure 1 shows the per-trial distribution (100 trials per diffusion time) of the relative score error $\|s^{\text{Gauss}} - s^{\text{grid}}\|_2 / \|s^{\text{grid}}\|_2$ of Gaussian projected BP on the Laplace chain.

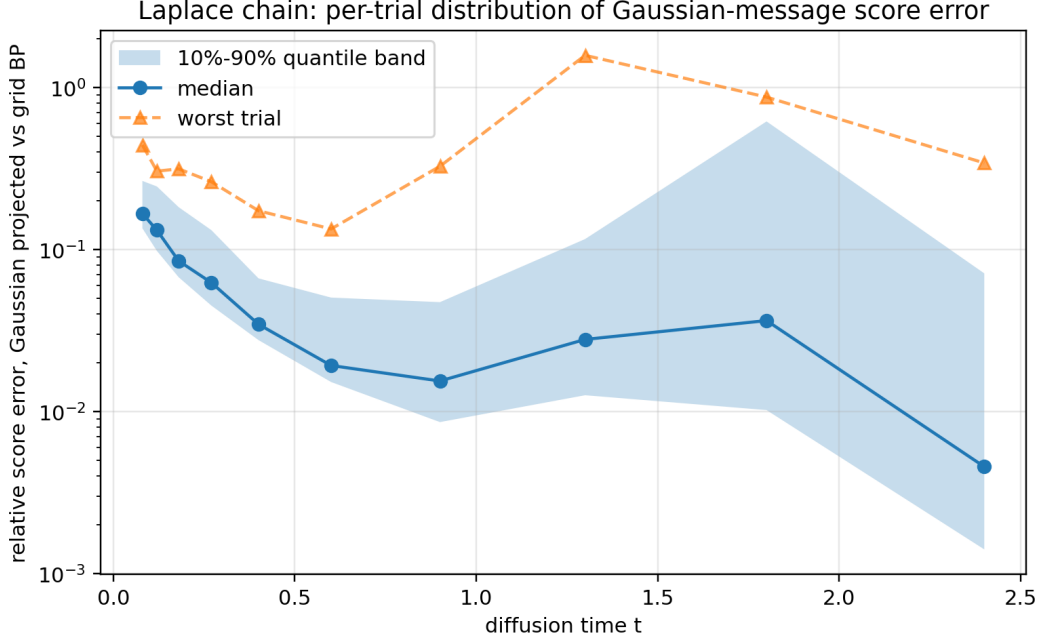


Figure 1: Per-trial distribution of the Gaussian-closure score error on the Laplace chain ($\rho = 0.85$, $n = 50$, reference grid $M = 401$). Small t : large, systematic error. Intermediate t : sub-percent. Large t ($t \gtrsim 1.3$): the widening band is dominated by a numerical instability of moment matching, see Section 5.

t	closure error (wide grid)	numerical floor [†]	median (100 trials)
0.08	0.211	10^{-15}	0.167
0.18	0.094	10^{-15}	0.085
0.40	0.044	2×10^{-7}	0.035
0.60	0.023	2×10^{-6}	0.019
0.90	0.010	3×10^{-4}	0.015
1.30	0.010	7×10^{-3}	0.028
1.80	—	0.16 (unreliable)	—

Table 1: Relative score error of Gaussian projected BP on the Laplace chain. “Wide grid” = mean over 16 trials on $[-12, 12]$. [†]The floor is the same algorithm run on the *Gaussian* chain, where projection is exact in principle, so any deviation is purely numerical. At $t \geq 1.8$ the floor exceeds the signal: those entries measure implementation fragility, not closure error.

Three regimes:

1. **Small t (near the data): the closure error is large and systematic.** At $t = 0.08$ the median relative score error is $\approx 17\%$ and *every one of 100 trials* exceeds 10%. This is not an outlier effect. The mechanism is the exact identity $s^{\text{Gauss}} - s^{\text{grid}} = (\alpha_t / \Delta_t) (m^{\text{grid}} - m^{\text{Gauss}})$ (verified to 10^{-14} in the runs): as $t \rightarrow 0$ the denoiser error is amplified by $\alpha_t / \Delta_t \sim 1/(2t)$, exactly where the posterior is most sensitive to the non-Gaussian innovation tails. The *direction* of the score survives better: cosine similarity ≥ 0.98 .

2. **Intermediate t : Gaussianization.** By $t \approx 0.9$ – 1.3 the closure error drops below 1%: the noisy marginal has effectively Gaussianized and Gaussian messages are adequate.
3. **Large t : an instability of naive moment matching (not closure error).**

5 The weak-information instability

The Gaussian control run exposes an important numerical phenomenon. For $t \gtrsim 1.8$ the likelihood precision $\alpha_t^2/\Delta_t \rightarrow 0$ and BP messages become nearly flat. Computing first and second moments of a nearly flat function on a finite grid is ill-conditioned, and the Gaussian projected algorithm develops rare but catastrophic failures — *even in the purely Gaussian model*, where projection should be exact (0.16 relative error at $t = 1.8$; widening the grid from $[-8, 8]$ to $[-12, 12]$ moves but does not remove the effect). On the Laplace chain at $t = 1.8$, 30% of trials exceed 10% score error for the same reason. Figure 2 shows the worst observed failure: the projected belief detaches completely from the true posterior mode.

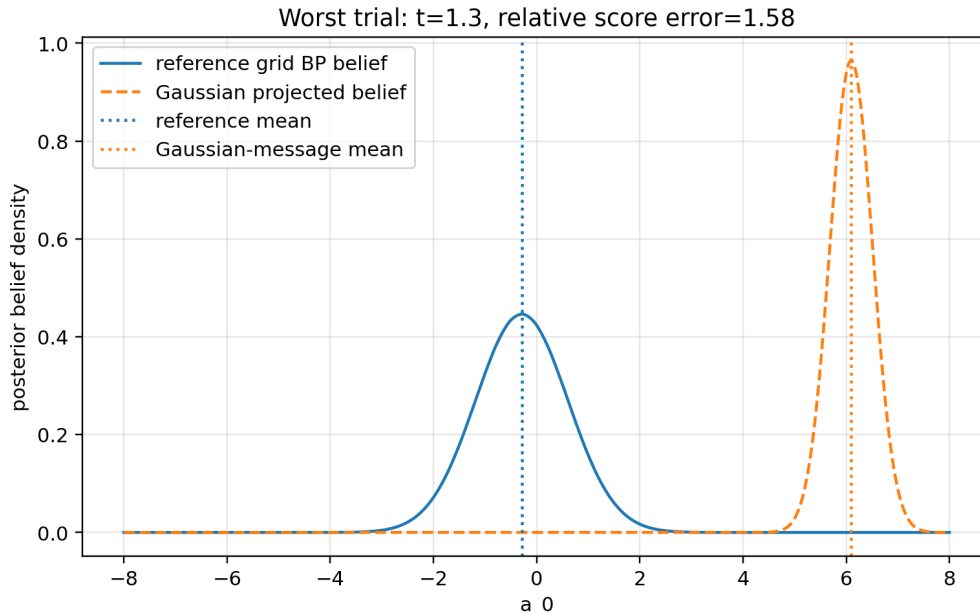


Figure 2: Worst failure over 1000 runs ($t = 1.3$): the moment-matched message chain places the belief at $a_0 \approx 6.1$ while the true posterior sits at -0.29 . A message-passing analogue of losing the mode.

The practical lesson: message *parametrization matters*. Moment matching on grids is fragile exactly in the weak-signal regime, whereas information-form (natural-parameter) updates — the ones used in Gaussian BP and in AMP derivations — handle flat messages gracefully ($\lambda \rightarrow 0$ is a perfectly regular point). This is an argument, from a purely numerical direction, for AMP-style parametrizations.

6 Proposed next steps

1. **Sharper small- t theory.** The measured closure error follows a clean monotone law in t . It should be possible to characterize the leading-order error in terms of the excess kurtosis of the innovations and the amplification factor α_t/Δ_t , and compare with the empirical curve.
2. **Beyond two moments.** Project messages onto richer families (Gaussian mixtures, or an Expectation-Propagation variant with damping in natural parameters) and measure how quickly the small- t gap closes as the family grows.
3. **Priors with real structure.** Replace Laplace innovations with sparse (spike-and-slab) or bimodal innovations, where the posterior genuinely multimodalizes and the speciation-like regime appears at finite t ; measure whether the closure error then develops a peak at intermediate t rather than decaying monotonically.
4. **Contact with AMP.** On dense graphs the same Gaussian-closure step is what turns BP into AMP. The chain setting gives a controlled laboratory where the exact answer is computable; the analogous comparison on dense/loopy models would quantify how much of AMP's error is Gaussian closure versus loop effects.

References

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- [2] M. Mézard and A. Montanari. Information, Physics, and Computation. Oxford University Press, 2009.
- [3] G. Genovese and A. Piana. Derivation of the AMP equations from belief propagation for the ℓ_2 minimisation problem. arXiv:2602.15191, 2026.